

Git & GitHub In-Class Assignment

1. Assignment Objective

The goal is to become comfortable with terminal commands, Git workflow, and pushing a local project to GitHub.

```
git --version
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
git config --global --list
git init
git status
git add .
git commit -m "message"
git log --oneline
git remote add origin <repo-url>
git branch -M main
git push -u origin main
git push
```

Important line to remember:

Commit is local. Push is online.

2. Tools Required

Before starting the assignment, students should have:

- Git installed
- GitHub account created

- VS Code installed
- Terminal available
- Internet connection

Recommended terminal:

- Windows: Git Bash, PowerShell, or VS Code terminal
 - macOS: Terminal or VS Code terminal
 - Linux: Terminal or VS Code terminal
-

3. Git Installation Guide

Windows

1. Open browser and search for **Git for Windows**.
2. Download Git from the official Git website.
3. Install Git with the default options.
4. After installation, open Git Bash, PowerShell, or VS Code terminal.
5. Verify installation:

```
git --version
```

Expected result:

```
git version x.x.x
```

macOS

Option 1: Install using Xcode Command Line Tools:

```
xcode-select --install
```

Option 2: Install using Homebrew:

```
brew install git
```

Verify installation:

```
git --version
```

Linux

For Ubuntu/Debian:

```
sudo apt update  
sudo apt install git
```

For Fedora:

```
sudo dnf install git
```

For Arch Linux:

```
sudo pacman -S git
```

Verify installation:

```
git --version
```

4. First-Time Git Configuration

Git needs your name and email to identify who made each commit.

Run these commands only once on your system:

```
git config --global user.name "Your Name"  
git config --global user.email "your-email@example.com"
```

```
git config --global --list
```

Example:

```
git config --global user.name "Rahul Kumar"  
git config --global user.email "rahul@example.com"  
git config --global --list
```

Important:

```
git config --global user.name "Your Name"
```

Do not write:

```
git config --global username "Your Name"
```

5. Quick Recap

Git

Git is a local version control tool. It helps us track code changes on our own computer.

GitHub

GitHub is an online platform where we can store Git repositories.

Repository

A repository is a project folder tracked by Git.

Working Directory

The folder where we edit files.

Staging Area

The place where we prepare files before commit.

Commit

A commit is a saved checkpoint in the local repository.

Push

Push uploads local commits to GitHub.

Main workflow:

```
Working Directory
  ↓ git add .
Staging Area
  ↓ git commit -m "message"
Local Repository
  ↓ git push
GitHub Repository
```

Daily workflow after first push:

```
Edit file → git status → git add . → git commit -m "message" → git push
```

6. Assignment Task

Task Name

Create and push a simple Git assignment project to GitHub.

Folder Name

Create one folder:

```
git-github-assignment
```

Inside that folder, create only these two files:

```
index.html  
README.md
```

7. Starter Code

index.html

Create a file named `index.html` and paste this code:

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title>Git Assignment</title>  
  </head>  
  <body>  
    <h1>Hello Git, please be kind today.</h1>  
  
    <p>My name is Your Name.</p>  
    <p>I am learning Git and GitHub.</p>  
  
    <h2>Today I will practice:</h2>  
    <ul>  
      <li>git init</li>  
      <li>git status</li>  
      <li>git add</li>  
      <li>git commit</li>  
      <li>git push</li>  
    </ul>  
  
    <p>Commit is local. Push is online.</p>
```

```
</body>
</html>
```

Change only this line:

```
<p>My name is Your Name.</p>
```

Replace `Your Name` with your actual name.

README.md

Create a file named `README.md` and paste this content:

```
# Git & GitHub Assignment
```

```
This is my first Git and GitHub assignment.
```

```
## About Me
```

```
My name is Your Name.
```

```
I am learning how to track code using Git and push code to GitHub.
```

```
## Commands I Practiced
```

```
- git init
- git status
- git add .
- git commit -m "message"
- git log --oneline
- git remote add origin <repo-url>
- git branch -M main
- git push -u origin main
- git push
```

```
## Important Line
```

Commit is local. Push is online.

Replace `Your Name` with your actual name.

8. Commands to Execute

Step 1: Open terminal inside the project folder

Make sure your terminal is opened inside:

```
git-github-assignment
```

Then run:

```
git status
```

If Git is not initialized yet, Git may show an error. That is okay.

Step 2: Start Git tracking

```
git init
```

Check status:

```
git status
```

Step 3: Add files to staging area

```
git add .
```

Check status again:

```
git status
```

Step 4: Create first commit

```
git commit -m "initial commit"
```

Check commit history:

```
git log --oneline
```

Step 5: Create GitHub repository

Go to GitHub and create a new repository.

Repository name:

```
git-github-assignment
```

Important: since we already created `README.md` locally, do not add README while creating the GitHub repository.

Do not select:

```
Add a README file  
Add .gitignore  
Choose a license
```

Create an empty repository.

Step 6: Connect local project to GitHub

Copy your GitHub repository URL.

Then run:

```
git remote add origin <repo-url>
```

Example:

```
git remote add origin https://github.com/your-username/git-github-assignment.git
```

Step 7: Push code to GitHub first time

```
git branch -M main  
git push -u origin main
```

Refresh your GitHub repository page.

You should see:

```
index.html  
README.md
```

Step 8: Make one small update

Open `index.html`.

Add this line before `</body>`:

```
<p>This is my second update. Git is slowly becoming my friend.</p>
```

Save the file.

Step 9: Push second update

```
git status  
git add .
```

```
git commit -m "added second update"
git push
```

Refresh GitHub again.

Your second update should be visible.

9. Final Self-Check

By the end of the assignment, check these points:

- I created a folder named `git-github-assignment`.
- I created `index.html`.
- I created `README.md`.
- I ran `git init`.
- I used `git status`.
- I used `git add .`.
- I created my first commit.
- I connected my local project to GitHub.
- I pushed my code to GitHub.
- I made one second update.
- I created one more commit.
- I pushed the second update to GitHub.
- I can see both files on GitHub.

Final line:

```
Commit is local. Push is online.
```